

Orange(Yingzhi) Ao

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Education

Nanjing University

B.A., Financial Engineering

2021-2025

Ranking: 3/29 | GPA: 4.60/5.0

Columbia University

Exchange Student

2024

GPA: 4.17/4.33

Relevant Courses

CS And Stats

**DS in Finance And Insurance [A+], **Predictive Modeling (in R), Statistical Machine Learning, Time Series, *Financial Engineering, *Artificial Intelligence, *Data Mining, *Options Investing Algorithms

Mathematics

**Stochastic Processes [rank 1], Discrete Mathematics [rank 1], Numerical Analysis (in MATLAB), ODE, *PDE, *Financial Mathematics

* Courses in progress.

** Grad-level courses.

Research Experiences

Multi-Period Compliance MFG-FBSDEs with DL

RA

2024.05 - Present

Supervisor: **Steven Campbell**

Columbia University

- **Innovation.** To solve **multi-period MFG** model for REC markets, I renovated **Deep FBSDE Solvers** and integrated numeric tricks for greater stability.
- **Results.** Please check out **the Report** and **my GitHub repository** for details. It will be made more realistic by adding more periods and/or inventory carryforward rules. (*Presented by Prof. Campbell in the 2024 INFORMS Meeting.*)

Insurance Pricing And Risk Modeling with ML

RA

2024.05 - 2024.08

Supervisors: **Carlos Arocha, Lina Xu**

Arocha Association, Columbia University

- **Innovation.** Trained **K-Means** to identify risk clusters and customized **exposure rating** models for each cluster to price the insurance policies, which have not been implemented by the industry.
- **Results.** Regularly delivered Statements of Work. Presented the strategy and ML techniques in a professional manner accessible to both technical and non-technical clients.

Low-carbon Transition of China's Power System

RA

2023.02 - 2023.08

Supervisor: **Yang Qu (Postdoctoral Fellow)**

Tsinghua-CTG Joint Center

- **Co-Authorship.** Co-authored the research report series "Low-Carbon Transition of China's Power System" and assisted in drafting and reviewing to uphold report quality.
- **Data Support.** Visualized data by plots/tables throughout the series to enhance statistical insights.

Project Experiences

Pairs Trading Optimization with GA

Self Directed

2023.08 - 2023.09

Applied **Genetic Algorithm** to optimize the spread weights in a cointegrated portfolio. Developed an enhanced Bollinger strategy with double bounds to improve timing for trading the optimized spread.

WorldQuant Brain

Consultant

2023.09 - 2023.12

Secured a gold medal (11th place out of 200+ participants) in the 2023 Brain x NJU Alphathon. Also contracted as a consultant to provide alpha ideas and received allowances.

Industrial Experiences

Higgs Assets Zhejiang, CHN	High-Freq Quant Researcher	2024.08 - Present
Developed a peer-grouping system to identify highly correlated CHN stocks using Affinity Propagation and XGBoost , which will be applied to boost the Multi-Head Attention Mechanism in the next phase.		
TrexQuant Stamford, CT (remote)	Global Alpha Researcher	2024.06 - 2024.08
Developed alphas for US stocks based on behavioral finance, incorporating sentiment, analysts, and technique data. Applied regression and clustering techniques.		
Chongwen Quantitative Beijing, CHN	Quant Strategy Researcher	2023.09 - 2024.02
Adapted and optimized the Whitney George Strategy to the CHN stock market. Built a trading simulation to generate instructions for traders. It's been actively used for trading since 2024 Q2.		

Skills

Programming

Proficient: PyTorch, Markdown, $\LaTeX$ Git commands. | Basic: R, C++, MATLAB.

Languages

English (TOEFL 111/120; GRE 333/340, with Quant 168/170). Mandarin (native).

Writing and Typesetting

Academic and professional reports, statements of work, and presentations.

Collaboration And Communication

Skilled in providing constructive feedback and adaptable in a team through teamwork experience.

Scholarships

• Jiangsu Huihong Livestock High Scholarship (5000 RMB)	2023.12
• Guotai Life Scholarship (2000 RMB)	2022.01
• People's Scholarship (2000 RMB)	2022.12